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Roll No

FT-801(A) (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Safety in Petrochemical Plants and Gas Terminals
(Elective - III)
Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.
ii) All questions carry equal marks.

1. Discuss the fire hazards, health hazards and dangerous properties of the following:

- i) Benzene
- ii) Ammonia
- iii) Naphtha

OR

Explain the hazards involved in the storage, handling and transportation of L.P.G.

2. Explain the material safety data sheet of gasoline based on flammability and reactivity data.

3. Write various advanced technologies used for fire fighting and safety in petrochemical plants.

4. Calculate the water requirement for the protection of oil tank wagon loading gantry based on data given below:

- i) Number of loading point = 70
 - ii) Length of rail gantry = 600m
 - iii) Width of tank wagon gantry = 12m
 - iv) C/C rail spacing = 7mtr
- Assume any missing data.

5. Explain in detail the following Refining process.

- a) Sulphur Recovery
- b) Polymerization

6. Discuss various fire fighting facilities at L.P.G. bottling plant in detail.

OR

Define and describe BLEVE. Also mention the fire fighting procedure in case of BLEVE.

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7.
 - a) Define Petroleum product? Give the detail classification of petroleum products?
 - b) Explain the Material Safety Data Sheet of CNG based on flammability and Reactivity data?
8. Answer any four of the following:
 - a) Write short note on petroleum classification.
 - b) Write a short note on hydro cracker.
 - c) Briefly describe various fire fighting installations in a refinery.
 - d) Describe fire fighting procedure for vehicles using LPG or CNG.
 - e) Write down basic provisions of Explosives Act 1984.
 - f) Describe in brief about the advanced technology “SCADA” used in refineries and petrochemical plants.
